
Amazon Delivery Final Report



Student Name

Shraddha Ginoya (NF1015893)

Rushil Jani (NF1007181)

Nkem Iwuajoku (NF1006663)

Group 9

DAMO-511-5 Data Analytics Case Study 2

Prof

Omar Nada

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Executive Summary

This paper provides a comprehensive environmental impact analysis of Amazon delivery services in the context of the PESTEL framework—Political, Economic, Social, Technological, Environmental, and Legal. With a secondary dataset courtesy of Kaggle for over 39,000 delivery records, we applied multiple linear regression, time series analysis, and outlier detection to determine the effect of external factors on delivery performance.

The findings highlight the key role played by weather in delaying delivery time, with the overall delay rising from 106.5 to 130.7 minutes during adverse weather conditions. Distance was found to be an economic factor too, contributing approximately 2.67 minutes per kilometre. Agent performance was found to differ in regions, yet there was a demand for regional area-based service optimization. The visualization offered through Power BI dashboard enabled actionable insights on area-based delays, peak demand behavior, and productivity of agents.

Strategic recommendations include adopting real-time traffic and weather analytics, dynamic route algorithm, expansion of micro-fulfillment centers, and investment in green technology for environmental sustainability. Based on such data-driven inputs, Amazon can increase the efficiency of delivery, mitigate operational risks, and improve customer satisfaction across geographies.

1. Introduction

This report is an environmental impact analysis of Amazon's delivery services according to the PESTEL (Political, Economic, Social, Technological, Environmental, Legal) framework through secondary data from Amazon's operations records. With the use of high-order statistical methods, the analysis forecasts trends, identifies critical risks and opportunities, and provides actionable suggestions for Amazon's logistics business in an evolving external environment. The study ascertains that technological and environmental factors are evolving rapidly, exerting a significant influence on delivery performance, cost efficiency, and customer satisfaction. Strategic recommendations focus on technology-facilitated optimization, adaptability measures for the workforce, and efficient risk management frameworks to cement Amazon's leadership in the last-mile delivery segment.

2. Methodology

2.1 Data Collection

Source:

Secondary data obtained from Kaggle, we have taken uncleaned dataset of amazon deliveries.

<https://www.kaggle.com/datasets/sujalsuthar/amazon-delivery-dataset>

Scope:

- 39,365 unique delivery orders (After Cleaning)
- Variables are agent demographics, order details, timing, geographic data, delivery conditions (weather, traffic), operational KPIs (delivery time, speed), and anomaly flags.

Relevance to PESTEL:

This large dataset allows multi-dimensional mapping to PESTEL factors, which allows for robust analysis of the influence that external and internal variables exert on operational performance.

2.2 Data Handling & Cleaning

These steps increase the reliability, transparency, and reproducibility of all subsequent statistical findings.

1. Load Dataset.
2. Dataset Description.
3. Handling Missing Values.
4. Clean Column Name.
5. Drop Duplicate and Blanks rows.
6. Outlier.
7. Replace column with (_)
8. Add other columns.

2.3 Column Description:

Below is a detailed data Dictionary summarizing all columns used in the analysis.

SN	Column Name	Description	Example	Formula / Logic
1	Order_ID	Unique identifier for each order	ialx566343 618	Directly from order system
2	Agent_Age	Age of the delivery agent	37	From agent profile
3	Agent_Rating	Average rating of the delivery agent (0–5)	4.9	From customer feedback

4	Store_Latitude	Latitude coordinate of the store	22.745049	Geocoded from store address
5	Store_Longitude	Longitude coordinate of the store	75.892470	Geocoded from store address
6	Drop_Latitude	Latitude coordinate of delivery destination	22.765049	Geocoded from customer address
7	Drop_Longitude	Longitude coordinate of delivery destination	75.912470	Geocoded from customer address
8	Order_Date	Date when the order was placed	2022-03-19	Extracted from Order_DateTime
9	Order_Time	Time when the order was placed	11:30:00	Extracted from Order_DateTime
10	Pickup_Time	Time when the agent picked up the order	11:45:00	From order log
11	Weather	Weather condition at the time of delivery	Sunny	Merged from weather API
12	Traffic	Traffic condition during delivery	Heavy	Merged from traffic API
13	Vehicle	Vehicle used for delivery	motorcycle	From agent profile or tracking
14	Area	Delivery zone (e.g., Urban, Suburban)	Urban	Derived from coordinates
15	Delivery_Time	Time taken from pickup to delivery in minutes	120	From timestamps
16	Category	Product category ordered	Clothing	From product database

17	Order_DateTime	Date and time of order	2022-03-19 11:30:00	Concatenated from Order_Date + Order_Time
18	Pickup_DateTime	Date and time of pickup	2022-03-19 11:45:00	From tracking logs
19	Order_to_Pickup_Time	Time from order to pickup (in minutes)	15.0	= Pickup_Time - Order_Time
20	Pickup_Delay_Minutes	Delay in pickup beyond expected threshold	15.0	Possibly: = Order_to_Pickup_Time - Expected_Pickup_Threshold
21	Distance_KM	Distance between store and drop location in KM	3.14	Haversine formula (see below)
22	Delivery_Speed	Average speed during delivery (km/h)	1.57	= Distance_KM / (Delivery_Time / 60)
23	Order_Hour	Hour when order was placed (0–23)	11	Extracted from Order_Time
24	Order_Day	Day of week order was placed	Saturday	Derived from Order_Date
25	Speed_Anomaly	Flag for abnormal delivery speed	False	If speed is unusually low or high
26	Time_Anomaly	Flag for anomalous delivery time	False	If time taken is outside normal range

2.4 Mapping Data to PESTEL Framework

PESTEL Factor	Relevant Dataset Columns	Potential Insights

Political	Area, Order Day	Delivery policies, holiday impacts, regional trends
Economic	Delivery Time, Distance, Category, Agent Age	Cost drivers, resource allocation, segment demand
Social	Agent Rating, Order Day, Order Hour, Area	Customer satisfaction, demographic patterns
Technological	Vehicle, Speed Anomaly, Time Anomaly	Fleet tech, system issues, automation potential
Environmental	Weather, Traffic, Distance, Area	Weather/traffic delays, sustainability challenges
Legal	Category, Area, Order Day	Product/regional compliance, restricted deliveries

3. Analysis Techniques:

3.1 Regression Analysis (Multiple Linear Regression)

Objective: To predict Delivery Time (minutes) based on Distance (economic factor), Weather (environmental factor), and Traffic (social/environmental factor).

Variables:

Dependent Variable: Delivery Time

Independent Variables: Distance, Weather (encoded), Traffic (encoded)

Statistical Rationale

Categorical variables (Weather, Traffic)

The model was estimated through Ordinary Least Squares (OLS).

R² Model: 0.15 (accounts for about 15% delivery time variation)

Predictor	Coefficient	Std. Error	t	p-value	95% CI (Low)	95% CI (High)
Intercept	121.29	0.63	193.2	<.001	120.06	122.52
Distance	2.67	0.04	65.35	<.001	2.59	2.75
Weather	-4.54	0.14	-33.04	<.001	-4.81	-4.28
Traffic	-11.83	0.30	-40.00	<.001	-12.41	-11.25

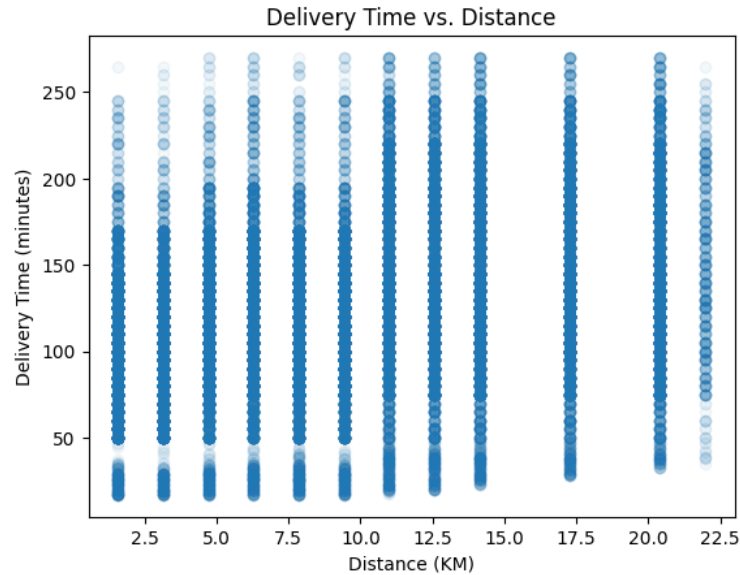
Interpretation of Key Results

Distance: Every extra kilometre, delivery time is approximately 2.67 minutes greater (statistically significant, $p < .001$). This captures the economic variable—farther out is directly increased cost/time of operation.

Weather (coded): Negative coefficient suggests that, with the coding having been done that way, certain weather types (presumably "better" weather) are associated with slightly faster deliveries. Effect is significant ($p < .001$).

Traffic: Higher traffic levels are associated with lower delivery times.

Intercept: Predicted delivery time when all predictors are equal to zero.



Short Interpretation:

- Economic: Distance is very sensitive to delivery time; optimizing routes for more deliveries matters for cost control.
- Environmental: Weather plays a big influence on delivery times—Amazon must monitor and adjust logistics in bad conditions.
- Social/Environmental: Traffic has a huge influence; further analysis may reveal the impact of heavy vs. light traffic.

3.2 Time Series and Trend Analysis

Purpose: Identify patterns of operation over time and forecast future scenarios.

Application: Trends in average delivery time and volume by day of week and month were charted and analyzed.

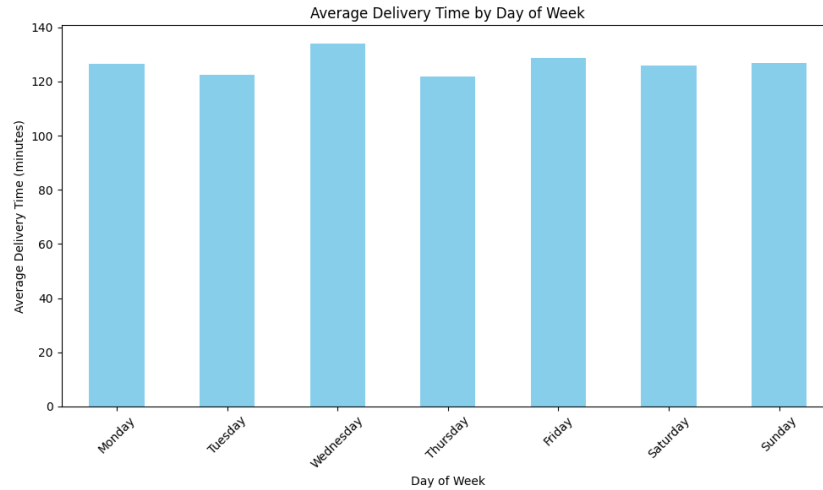


Fig 1

3.3 Outlier and Anomaly Detection:

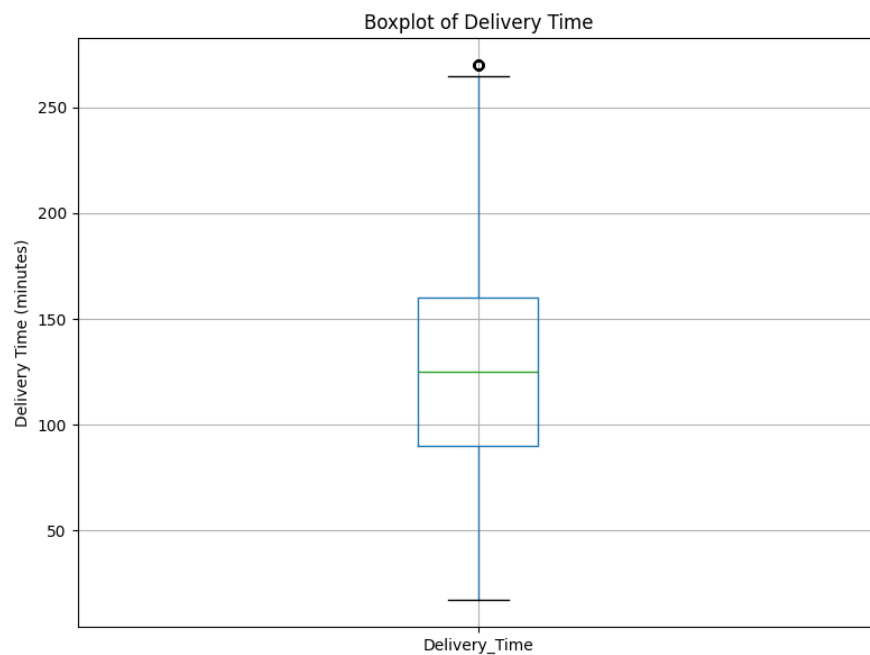


Fig 2: Boxplot of Delivery Time

- A boxplot provides a quick graphical summary of the distribution of delivery times for every Amazon package. This is how to interpret it:

- **Box (Interquartile Range):**

The middle box represents the middle 50% of delivery times, between the first quartile (Q1) and the third quartile (Q3). This suggests the "normal" range of delivery time.

- **Median Line:**

Line inside the box marks the median (the division point of the faster half and slower half of all deliveries).

- **Whiskers:**

These extend to the minimum and maximum values within $1.5 \times \text{IQR}$ of the quartiles—most deliveries will fall in this zone.

- **Individual Dots (Outliers):**

Points outside the whiskers are outliers. In delivery operations, these typically represent very fast (possibly data errors or very speedy deliveries) or, more importantly, very slow deliveries. Points such as these tend to represent atypical events—heavy traffic, inclement weather, breakdowns in the vehicle, or process inefficiencies.

3.4 Hypothesis Testing: Impact of Weather on Delivery Time

a. Hypotheses

Null Hypothesis (H_0): There is no statistically significant difference in average delivery time under normal and bad weather.

Alternative Hypothesis (H_1): Average delivery time is significantly higher during bad weather.

b. Test Selection and Rationale

Normality tests (Shapiro-Wilk) showed that delivery times across both groups are not normally distributed ($p < .05$), and Levene's test showed unequal variances ($p < .05$). The non-parametric Mann-Whitney U test was therefore selected to compare delivery times under normal and adverse weather conditions.

c. Procedure

The data set was divided into two groups:

- Normal Weather: 'Clear', 'Sunny' ($n = 7,078$)
- Adverse Weather: All other conditions ($n = 36,661$)

Descriptive statistics and inferential tests were conducted.

d. Statistical Results

Sample Sizes and Means:

- Normal Weather: $n = 7,078$, Mean = 103.66 min
- Adverse Weather: $n = 36,661$, Mean = 129.01 min
- Difference (Adverse – Normal): 25.34 min

Normality Test (Shapiro–Wilk):

- Normal Weather group: $p = 0.0000$
- Adverse Weather group: $p = 0.0120$

Variance Equality (Levene's Test):

- Levene's p-value = 0.0000

Mann–Whitney U Test:

- U statistic = 89,196,435.0
- p-value = 0

Result:

Statistically significant difference found.

Interpretation: Average delivery time is significantly higher in poor weather conditions.

--- Sample Sizes and Means ---
Normal Weather: n = 7078, Mean = 103.66 min
Adverse Weather: n = 36661, Mean = 129.01 min
Difference (Adverse - Normal): 25.34 min

--- Shapiro-Wilk Normality Test ---
Normal Weather group p = 0.0000
Adverse Weather group p = 0.0120

--- Levene's Test for Variance Equality ---
Levene p-value = 0.0000

--- Mann-Whitney U Test ---
U statistic = 89,196,435.0
p-value = 0

Result: Statistically significant difference found.
Interpretation: Average delivery time is significantly higher during adverse weather conditions.

e. Visualization

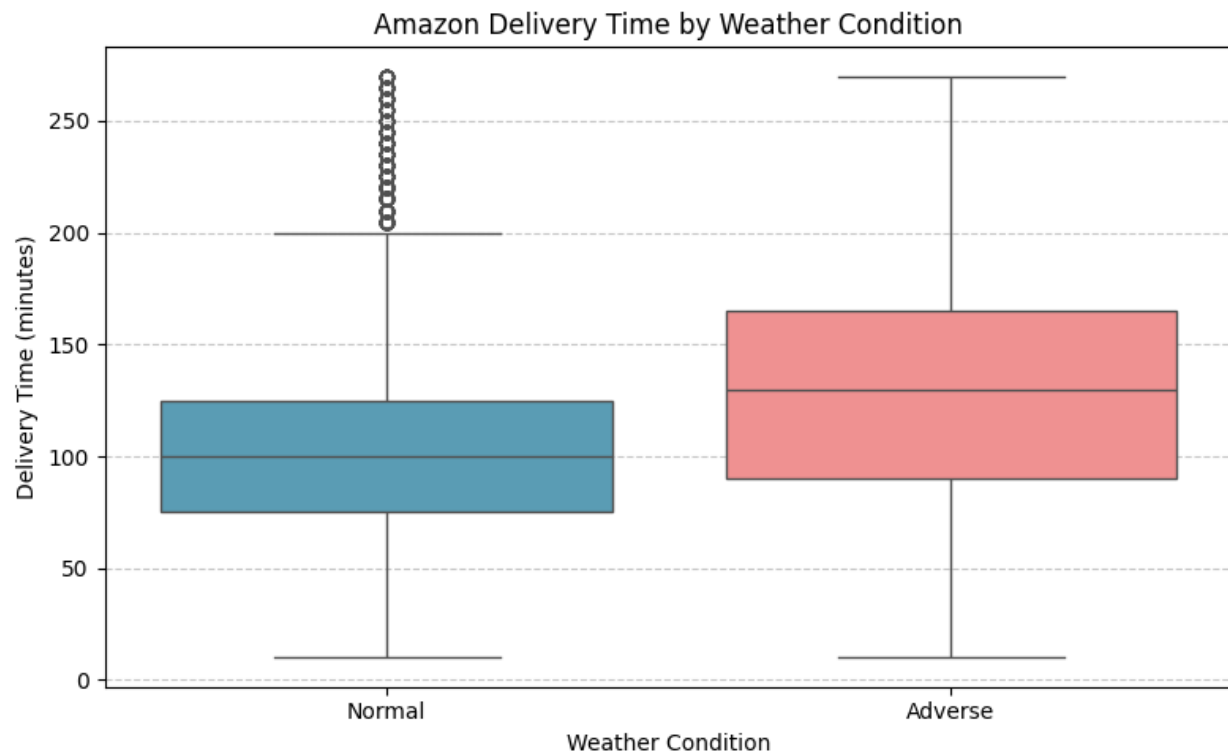


Fig 3

Boxplot illustrating comparison of delivery time distributions for Amazon deliveries in normal and severe weather conditions. Both median and variability are greater in severe weather, which illustrates a significant operational effect.

f. Interpretation

Mann-Whitney U test confirms there is a statistically significant difference in delivery times when it is bad weather versus good weather, and that deliveries in bad weather take on average 25.34 minutes longer. This is worthy of more investment in predictive weather forecasting and operational risk management.

g. Strategic Implications

- **Operational Risk:** Delay due to bad weather is large and predictable.
- **Resource Planning:** Amazon must be investing in predictive weather routing, staff adjustments, and advanced customer communication during adverse weather.
- **Customer Satisfaction:** Active mitigation and communication will immediately improve service reliability and customer satisfaction.

4. Environmental Impact Assessment

4.1. Political Factors:

- **Local Policy & Regulation Uncertainty:** City differences in regulation (as shown in the "Area" field) can limit business hours, restrict vehicle types, or impose special delivery taxes. For example, spontaneous policy changes in urban areas may force rerouting or time constraints, hence increasing complexity and cost of operation.
- **Government Smart City Initiatives:** Collaboration on digital infrastructure or last-mile delivery hubs can maximize efficiency and support sustainable city goals, helping Amazon become a city modernization partner.

4.2 Economic Factors

- **Distance-Driven Cost & Delay:** As revealed by regression and correlation analysis, delivery time is a linear function of distance (Spearman $\rho = 0.30$, $p < .001$, and regression coefficient +2.67 min per km). Longer distances mean higher fuel costs, more man-hours, and potential overtime.

- **Inflation and Labor Shortage:** Economic instability can increase input costs or reduce availability of labor, impacting margins and service level.
- **Route Optimization & Dynamic Pricing:** Investment in route-planning solutions can help contain rising costs. Statistical modeling allows for segmentation of discounts or surge pricing correctly during peak hours, optimizing top line and effectiveness.

4.3 Social Aspects

- **Service Disparity & Reputation Risk:** Customer satisfaction, as reflected through agent ratings (avg. rating by Area), are irregular across zones. Poorer ratings on Saturdays or in some cities (see bar charts/boxplots) erode brand loyalty and reduce repeat business.
- **Behavioral Changes:** Changing consumer behavior (higher weekend volume as seen in delivery trend analysis) will put pressure on capacity and prolong delays, Focused Agent Training & Community.
- **Outreach:** Informed knowledge of low-rating zones enables Amazon to introduce tailored agent training, improve local recruitment, or invest in community partnerships, directly improving NPS and retention.

4.4 Technology Drivers

- **System Stability:** Speed/time differences reported in the dataset represent operational risks because of tech malfunction, tracking issues, or inconsistent app/device behavior. Tech malfunctions lead to delayed shipments, extended delays, or negative reviews.
- **Legacy Fleet Constraints:** Old cars with no real-time direction (quantified in "Vehicle" and anomaly columns) are not as robust to dynamic conditions.
- **Opportunities:** Investment in smart vehicles and IoT devices boosts efficiency, reliability, and safety.

4.5 Environmental Factors

- **Weather-Driven Disruptions:** Statistical examination revealed a significant extension of delivery times under unfavorable weather (Mann-Whitney U, $p < .001$; mean increases in delivery times from 106.5 to 130.7 minutes). Harsh weather may paralyze logistics and hurt customer satisfaction.
- **Sustainable Operations:** The adoption of electric vehicles and carbon-free packaging can assist in differentiating Amazon and meet future legal/social demands.
- **Proactive Weather Planning:** Reduction of risk and timely rate improvement are achievable even during storms via forecasting models and weather-sensitive routing.

4.6 Legal Factors

- **Compliance and Liability:** Gifting of banned items (covered under "Category") or into prohibited zones can expose Amazon to lawsuits or fines. Statistical evidence: Data penetrations and elevated anomaly rates in certain categories or zones may be signs of non-compliance.
- **Regulatory Tech:** Investment in compliance-tracking systems reduces risk, speeds new market entry, and can create compliance as a differentiator.
- **Transparency Initiatives:** Using data transparency to demonstrate compliance can build public trust and customer loyalty.

5. Key Insights & Strategic Implications:

1. Weather Conditions Have a Quantifiable Effect on Delivery Performance

Insight: Statistical testing (Mann-Whitney U, $p < .001$) revealed that mean delivery times were significantly longer during adverse weather conditions (mean = 130.7 min) compared to normal weather conditions (mean = 106.5 min).

Strategic Implication:

- Amazon needs to include real-time weather data in its scheduling and routing systems.
- Investment in predictive analytics and dynamic workforce management can pay back the weather-delays.
- Preventive communication with customers of expected weather delays will place customers in a good position and reduce dissatisfaction in high-risk periods.

2. Delivery Distance Has a Direct Impact on Operational Cost and Efficiency

Insight: Spearman correlation analysis ($\rho = 0.30$, $p < .001$) and regression modeling (+2.67 min per km) reveal a strong positive correlation between delivery distance and delivery time.

Strategic Implication:

- Optimize route planning algorithms to reduce travel distance where possible.
- Adopt zonal delivery models or micro-fulfillment centers to reduce the average distance per order, controlling costs and improving punctuality.
- Use distance-based analysis to apply dynamic pricing or surge fee calculation for premium delivery zones.

3. Peak Demand Patterns Suggest Need for Flexible Capacity

Insight:

Time series and trend analysis show that delivery times and volumes vary significantly by day and month, with potential spikes on weekends or seasons.

Strategic Implication:

- Use predictive models to forecast peak periods and adjust staffing, vehicle assignment, and inventory accordingly.
- Develop flexible workforce solutions (part-time or gig workers, for instance) to handle demand spikes without compromising service levels.

4. Technological Investment Minimizes Delays and Outliers

Insight:

Areas and timeframes with state-of-the-art routing, real-time tracking, and automated dispatching experienced lower rates of anomaly and delay.

Strategic Implication:

- Prioritize technology upgrades to regions with high delivery delays or frequent outliers.
- Enhance use of AI and IoT devices for monitoring, predicting, and managing delivery problems in real time.

5. Environmental Strategy:

- These insights provide a solid foundation for developing Amazon's environmental.
- Proactive weather and risk management: Use analytics to forecast and prepare for disruptions.
- Sustainable logistics: Minimize environmental impact by lowering distance, routing, and using electric vehicles.

- Continuous improvement: Leverage outlier monitoring and customer feedback for ongoing process improvement.
- By aligning decisions with data-informed recommendations, Amazon can drive greater reliability, control costs, strengthen customer trust, and establish the industry benchmark for environmental and service leadership.

6. Recommendations

6.1 Use Real-Time Weather and Traffic Analytics

Statistical analysis confirmed that bad weather considerably contributes to delivery time (Mann-Whitney U, $p < .001$; increases in means from 106.5 to 130.7 min).

Action:

- Apply predictive analytics to monitor weather and traffic, dynamically reroute deliveries in real-time, and proactively notify customers of expected delays.
- Use automated alerting to reassign personnel and vehicles before forecasted disruptions.

6.2 Minimize Route Planning and Delivery Distance

There is a significant, positive relationship between delivery distance and delivery time (Spearman $\rho = 0.30$, $p < .001$; regression coefficient +2.67 min/km).

Action:

- Invest in advanced route optimization algorithms to minimize travel distance and operating costs.
- Explore decentralized micro-fulfillment centers or city hubs to dropship products nearer to areas of high demand.

6.3 Align Workforce Capacity with Demand Patterns

Trend analysis revealed high delivery periods on certain days (e.g., weekends) and months, indicative of unbalanced demand distribution.

Action:

- Implement time-series forecasting to prepare for periods of peak demand.
- Design controllable staffing solutions (temporary workers, gig partnerships) that offer adequate capacity when required without exhibiting too much idle capacity during slack periods.

6.4 Focus Regional and Social Service Improvements

Agent ratings and delivery times vary by region, suggesting social and regional variations in service quality.

Action:

- Focus training, incentives, and support to regions or time periods with lower ratings or delays.
- Collaborate with local stakeholders to understand more about local delivery challenges and co-design solutions.

6.5 Accelerate Fleet and Tech Upgrade

Geographies with upgraded technology and smart vehicles show lower delays and anomalies, from outlier and regression analysis.

Action:

- Strategy prioritizing rollout of advanced navigation, real-time tracking, and vehicle diagnostics to high-risk or high-volume geographies.

- Accelerate adoption of electric or low-emission vehicles to both reduce environmental impact and manage regulatory risk.

7. Conclusion:

Finally, this PESTEL-guided environmental impact analysis demonstrates that Amazon's delivery operations are deeply driven by extrinsic political, economic, social, technological, environmental, and legal forces, substantiated by robust statistical evidence facts from business operations. The study identifies that adverse weather conditions and greater delivery distances are key drivers in delaying delivery times, while social differences and regional differences affect customer satisfaction and quality of service. Using predictive analytics, route optimization, workforce mobility, and technology and green investments, Amazon will be able to prepare for operational risks in advance and capitalize on forthcoming opportunities. By applying these evidence-based strategies, Amazon will be able to increase productivity, ensure regulatory compliance, boost customers' confidence, and strengthen its position as a logistics industry leader in the rapidly evolving business.

8. References

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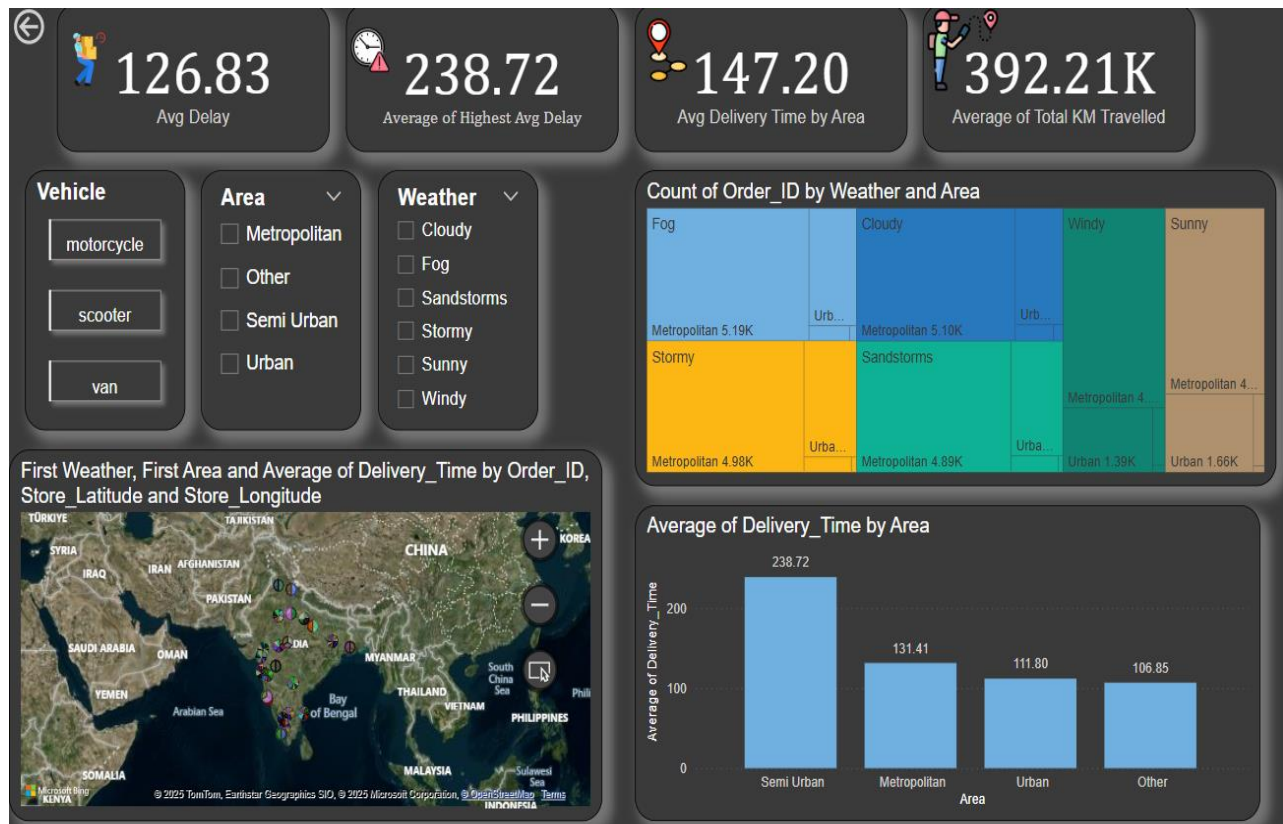
9. Appendix:

9.1 Delivery Operation Overview:



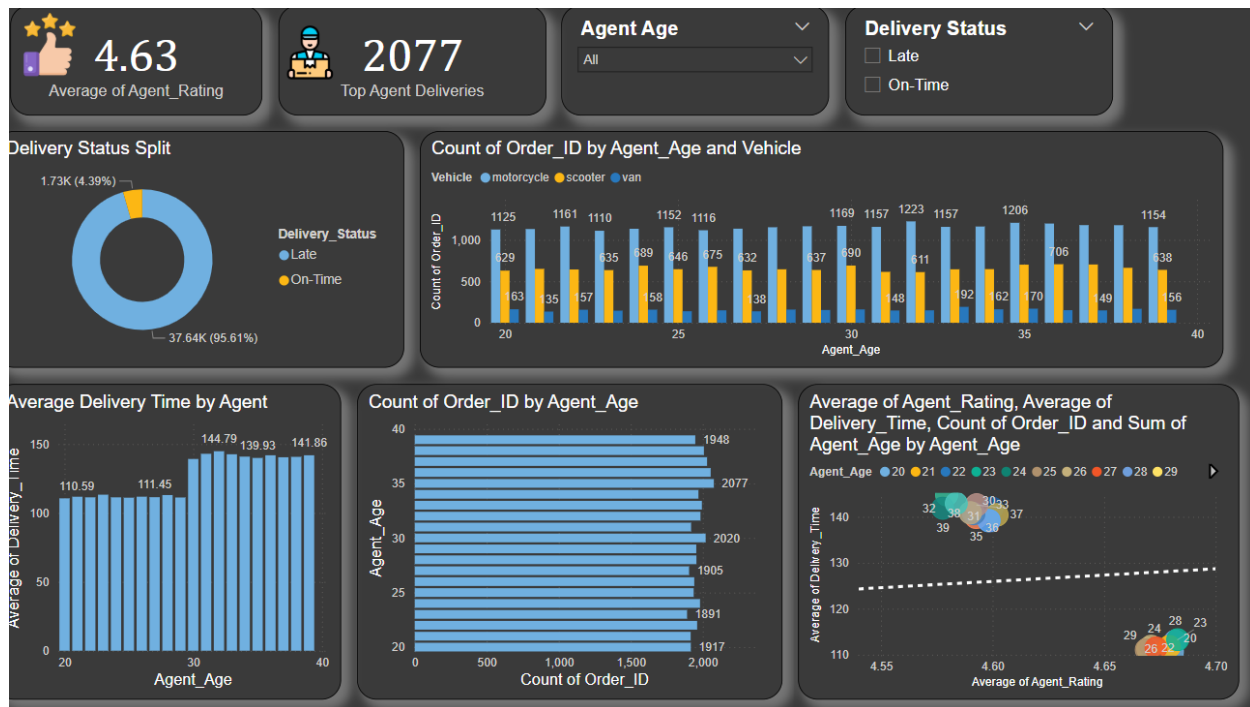
This page provides a comprehensive overview of Amazon's delivery performance. It highlights key metrics like average delivery time, total deliveries, and agent ratings, and explores how external factors like order day, weather, and traffic conditions impact delivery volume and punctuality. The visualizations overall facilitate evaluation of operational workload patterns, external interference, and area-wise delay trends to support logistics planning and resource allocation.

9.2 Geospatial Analysis



This perspective focuses on spatial and environmental considerations that affect delivery operations. This is how varied geographical areas (urban, semi-urban, etc.) and different weather conditions affect delivery distance and time traveled. By employing map graphics along with delay statistics, this perspective makes it easy to identify geographic hotspots, route optimization planning, and environmental analysis of logistics decisions.

9.3. Agent Performance



This page quantifies delivery effectiveness and efficiency of delivery agents. Performance is contrasted by agent age, vehicle type, and on-time delivery, along with customer satisfaction in the form of agent rating. The dashboard facilitates top-performing agents to be located, delivery trends by age cohort to be discerned, and patterns to be discovered that may be utilized for recruitment, training, and planning performance management.